Summary in Graph

Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 6)

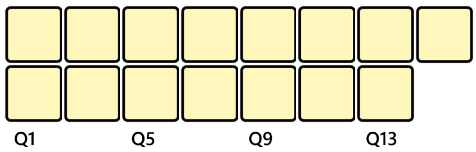
Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Q12). Link
Type: Multiple Select
Topic: goclasses,principal-component-analysis
Marks Obtained: 0
Time Taken: 00min 00sec
Avg. Time Taken: 03min 10sec
Avg. Accuracy: 70 %
Avg. Skipped: 3 %

Individual Question Statis

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (33) out of 331 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Why is PCA sometimes used as a preprocessing step before regression?

A. To reduce overfitting by removing poorly predictive dimensions.
B. To expose information missing from the input data.
C. To make computation faster by reducing the dimensionality of the data.
D. For inference and scientific discovery, we prefer features that are not axis-aligned.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 14sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Select the true statements about principal components analysis (PCA).

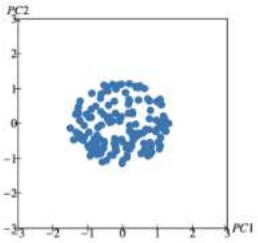
A. PCA is a clustering algorithm.
B. PCA produces features (principal coordinates) that are linear combinations of the input features.
C. The principal components are chosen to maximize the variance in the projected data.
D. The principal coordinates are the eigenvalues of the sample covariance matrix.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

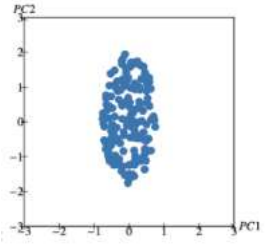
Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

In each plot below, the data is projected onto two (unit-length) principal component vectors. We say that a plot is "valid" if the x-coordinate (written as "PC1") would be the first chosen principal coordinate and the y-coordinate (written as "PC2") would be the second chosen principal coordinate. Which plots are valid?

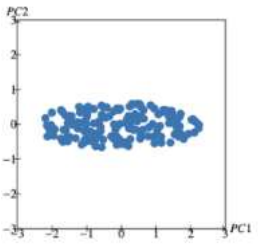
A.



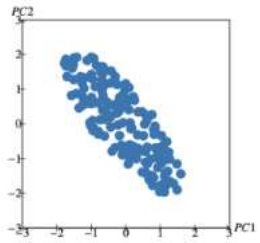
B.



C.



D.



Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following three points in 2-dimensional space: $(1,1),(2,2),(3,3)$.
Suppose we project the original data points into a 1-dimensional space defined by the first principal component.
The variance of the projected data is:

A. $2/3$
B. $4/3$
C. $1/3$
D. 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider n feature vectors $\boldsymbol{x} \in \mathbb{R}^d$ that need to be projected into a lower-dimensional space $\mathbb{R}^k$ using k principal component basis vectors. The projection matrix $\boldsymbol{W}$ of size $d \times k$ consists of these orthonormal principal component vectors.
Which of the following expressions correctly represents the reconstruction error?

A. $\left\|\boldsymbol{x} - \boldsymbol{W}\boldsymbol{W}^T\boldsymbol{x}\right\|_2^2$
B. $\left\|\boldsymbol{x} - \boldsymbol{W}^T\boldsymbol{x}\right\|_2^2$
C. $\left\|\boldsymbol{x} - \boldsymbol{W}^T\boldsymbol{W}\boldsymbol{x}\right\|_2^2$
D. $\left\|\boldsymbol{x} - \boldsymbol{W}\boldsymbol{x}\right\|_2^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Given a data point $\boldsymbol{x}$ and a unit vector $\boldsymbol{v}$ along the principal axis of the projection, the projected vector $\boldsymbol{x}_{\|}$ is defined as:

$$\boldsymbol{x}_{\|} = \boldsymbol{v}\boldsymbol{v}^T\boldsymbol{x}$$

The difference vector between the data point and its projection is:

$$\boldsymbol{x}_{\perp} = \boldsymbol{x} - \boldsymbol{x}_{\|}$$

Which of the following statements is/are correct?

A. $\boldsymbol{x}_{\perp}$ and $\boldsymbol{v}$ are orthogonal.
B. $\boldsymbol{x}_{\perp}$ and $\boldsymbol{v}$ are parallel.
C. $\boldsymbol{x}$ and $\boldsymbol{x}_{\|}$ are orthogonal.
D. $\boldsymbol{x}$ and $\boldsymbol{x}_{\|}$ are parallel.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Assume that we have the following covariance matrix XX^T, please calculate the first principal component.

$$XX^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

A. $[1, 0, 0]^T$
B. $[0, 1, 0]^T$
C. $[0, 0, 1]^T$
D. $[1, 2, 3]^T$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose the first six singular values are 6, 5, 3, 2, 2, 1 .
What fraction of the variance is captured by the first two principal components?

A. $\frac{6^2}{6^2+5^2+3^2+2^2+2^2+1^2}$
B. $\frac{6^2+5^2}{6^2+5^2+3^2+2^2+2^2+1^2}$
C. $\frac{6+5}{6+5+3+2+2+1}$
D. $\frac{6^2+5^2+3^2}{6^2+5^2+3^2+2^2+2^2+1^2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose you have four observed data vectors, each of which is a multiple of the 6-dimensional vector $\vec{v}$ with $\|\vec{v}\| = 10$. You arrange these data vectors as the columns of a matrix A given by:

$$A = \begin{bmatrix} | & | & | & | \\ -\vec{v} & -2\vec{v} & \vec{v} & 2\vec{v} \\ | & | & | & | \end{bmatrix}$$

Find the singular value that corresponds to the first (and only) principal component.

A. 10
B. $10\sqrt{2}$
C. $10\sqrt{5}$
D. $10\sqrt{10}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Suppose we perform PCA on a dataset in order to reduce the dimensionality to 3 . Let Z be the data represented in the new feature representation, after performing PCA. Suppose the covariance matrix of Z is computed. Which of these could possibly be the covariance matrix? Select all that apply.

A. $\begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
B. $\begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

C. $\begin{pmatrix} 6 & -1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

D. $\begin{pmatrix} 6 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Which of the following are true about principal components analysis (PCA)? Assume that no two eigenvectors of the sample covariance matrix have the same eigenvalue.

A. Appending a 1 to the end of every sample point doesn't change the results of performing PCA (except that the useful principal component vectors have an extra 0 at the end, and there's one extra useless component with eigenvalue zero).
B. If you use PCA to project d-dimensional points down to j principal coordinates, and then you run PCA again to project those j-dimensional coordinates down to k principal coordinates, with $d > j > k$, you always get the same result as if you had just used PCA to project the d-dimensional points directly down to k principle coordinates.
C. If you perform an arbitrary rigid rotation of the sample points as a group in feature space before performing PCA, the principal component directions do not change.
D. If you perform an arbitrary rigid rotation of the sample points as a group in feature space before performing PCA, the largest eigenvalue of the sample covariance matrix does not change.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Suppose we are doing ordinary least-squares linear regression with a fictitious dimension. Which of the following changes can never make the cost function's value on the training data smaller?
A. Discard the fictitious dimension (i.e., don't append a 1 to every sample point).
B. Append quadratic features to each sample point.
C. Project the sample points onto a lower-dimensional subspace with PCA (without changing the labels) and perform regression on the projected points.

D: Center the design matrix (so each feature has mean zero).

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Which of the following are typical benefits that might motivate you to preprocess data with principal components analysis (PCA) before training a classifier?

A. PCA tends to reduce the bias of your classification algorithm.
B. PCA can be used to avoid overfitting.
C. PCA tends to reduce the variance of your classification algorithm.
D. PCA can be used to avoid underfitting.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

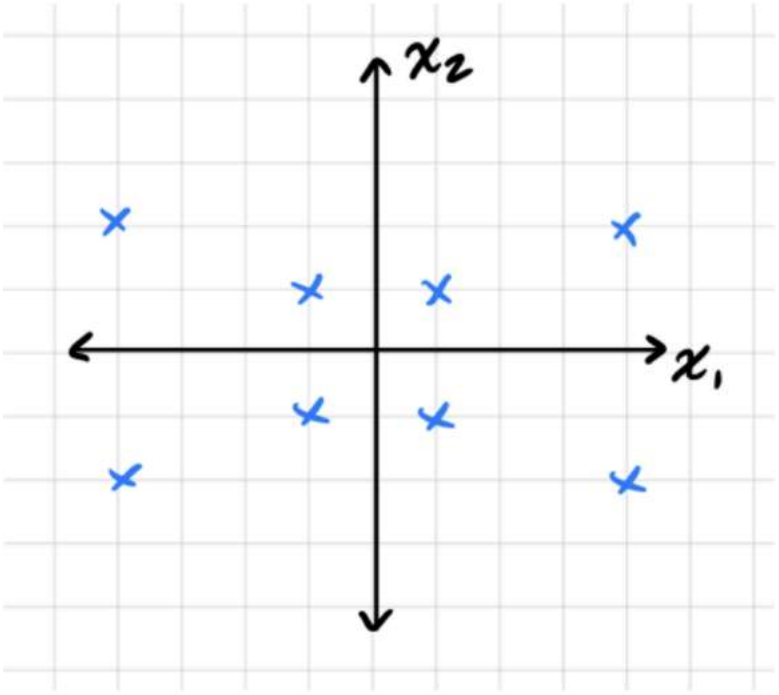
You performed PCA in $\mathbb{R}^3$. The first principal component is $u_1 = (1/\sqrt{2}, 1/\sqrt{2}, 0)$, and the second principal component is $u_2 = (-1/\sqrt{2}, 1/\sqrt{2}, 0)$. One of your original data points was $x = (1, 2, 3)$.
What will the new representation of x be?

A. $\left(\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$
B. $\left(\frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
C. $\left(\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$
D. $\left(-\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following data:



Assume every sample data point is marked with "x" and has integer coordinates only. If we project the data onto the first principal component, what will be the reconstruction error?

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

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